# Virtual Machines vs Containers

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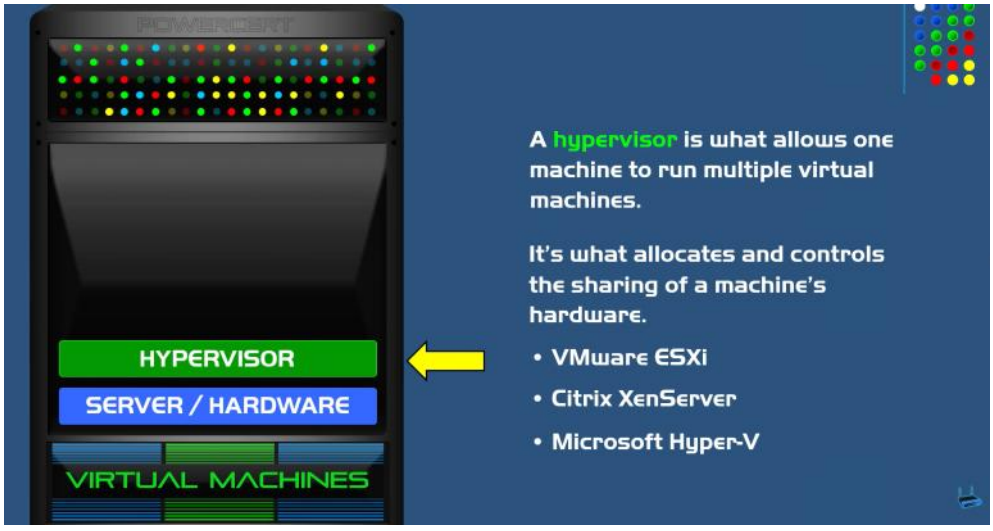
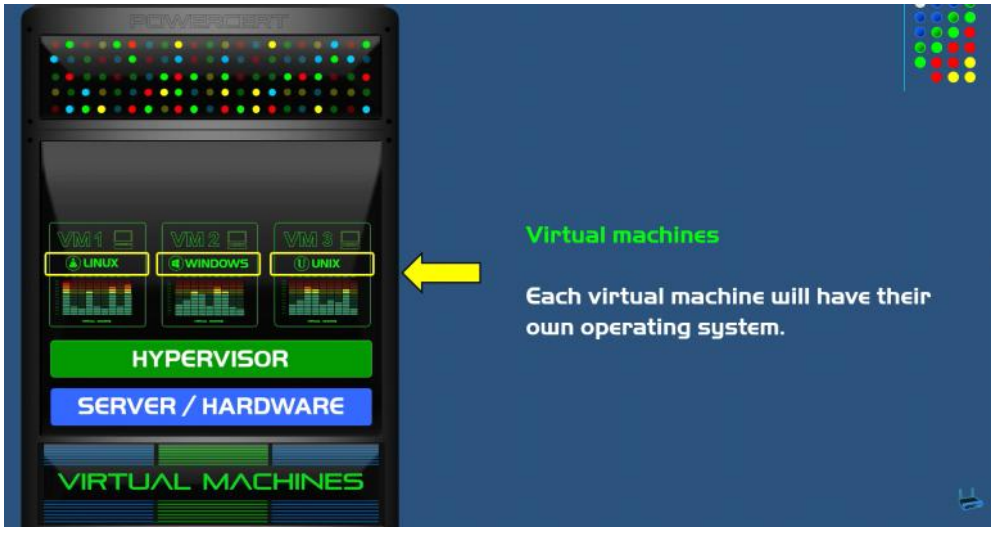
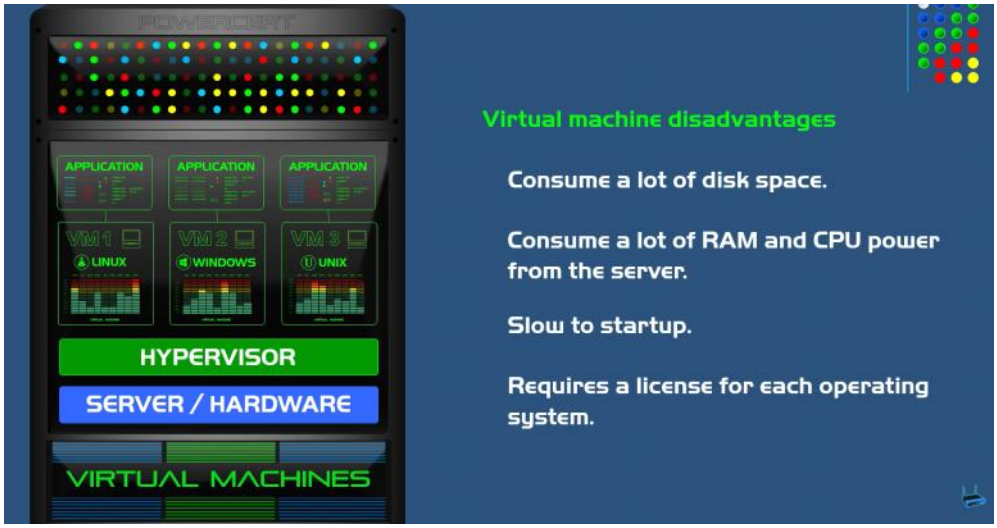
**Virtual Machines vs Containers**

**Traditional Approach**
Previously, businesses ran only one application per server. This was inefficient because servers were often underutilized, leading to wasted hardware capacity and unnecessary costs.

**Virtual Machines (VMs)**
VMs solve this by using a hypervisor to simulate multiple virtual machines on a single physical server. Each VM includes its own full operating system, allowing multiple applications to run independently. While effective, VMs are resource-intensive, consume significant disk space/RAM, and are slower to boot because they must load an entire OS.

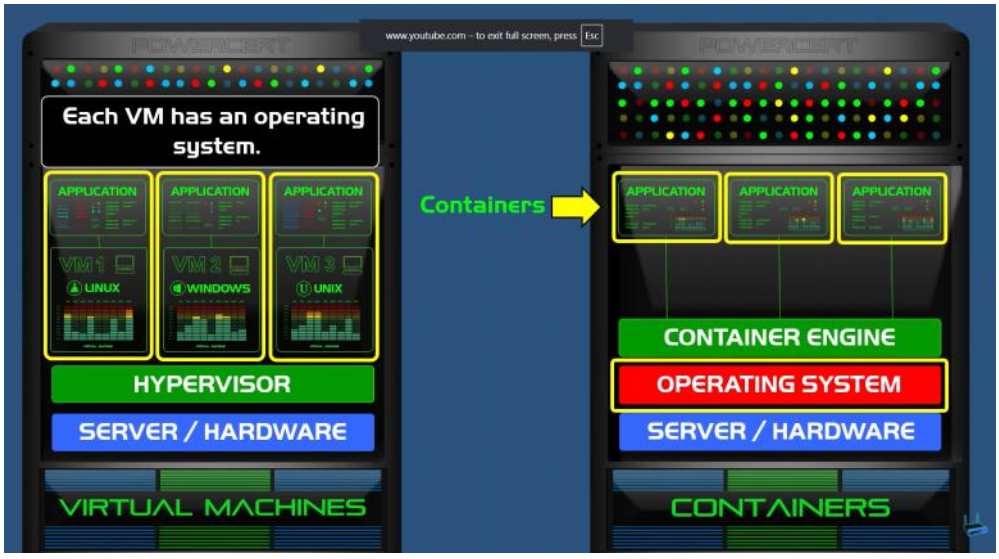





**Containers**

Containers focus on bundling just the application, along with its necessary files, configurations, and dependencies. They share the host server's operating system kernel, making them lightweight, fast to boot (milliseconds), and highly portable. Docker is the leading tool for managing these packages.




**Key Differences**

- Isolation: VMs provide full OS-level isolation, whereas containers share the host OS, making them more efficient but slightly less isolated.
- Performance: Containers are much smaller and faster than VMs.
- Compatibility: VMs can run any operating system, while containers must be built to match the host server's OS (e.g., Linux containers for a Linux host).